## Investigation: RECRUSUL S.A. (CVM code 12572), fiscal year 2014, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
Before 2018, the company was called RECRUSUL SA.
Trading codes (2023 FCA, version 1, filed on 2023-05-08): RCSL3 (Ações Ordinárias, Bolsa, since 1985-12-20); RCSL4 (Ações Preferenciais, Bolsa, since 1985-12-20).

## Document used

2014 DFP, version 1, filed on 2015-06-30; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2015-06-30).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=48877

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −9.616 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 8 (n = 18)
- Percentile in group and year: 1 (n = 374)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.599 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 89 (n = 22)
- Percentile in group and year: 94 (n = 464)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.218 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 93 (n = 22)
- Percentile in group and year: 87 (n = 483)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.227 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 22)
- Percentile in group and year: 1 (n = 473)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.108 (index)
- M-score component; it has no rule of its own

Percentile in sector and year: 7 (n = 22)
Percentile in group and year: 1 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.040 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 64 (n = 18)
Percentile in group and year: 74 (n = 394)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 10.876 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 22)
Percentile in group and year: 99 (n = 477)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.967 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 11 (n = 22)
Percentile in group and year: 2 (n = 503)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 2.460 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 22)
Percentile in group and year: 98 (n = 492)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 1,179.8 days
Alert by the study 1 rule: yes
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 27.9 days)
Percentile in sector and year: 92 (n = 20)
Percentile in group and year: 99 (n = 447)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 325.55 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 93 (n = 21)
Percentile in group and year: 98 (n = 442)

The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.536 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 11 (n = 22)
Percentile in group and year: 3 (n = 496)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −30.969 (index)
Components: X1 = −1.770; X2 = −4.455; X3 = −0.610; X4 = −0.700 (indices)
Percentile in sector and year: 11 (n = 23)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 4 accounts above the R$1,000 tolerance (group a, totals and key lines: 0; group b, reclassifications: 4), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-29
- DFP 2011, v1: 2012-03-30

- DFP 2012, v1: 2013-03-27
- DFP 2012, v2: 2013-04-04
- DFP 2013, v1: 2014-03-31
- FCA 2014, v1: 2014-03-31
- FCA 2014, v2: 2014-05-13
- DFP 2014, v1: 2015-06-30
- DFP 2015, v1: 2016-08-15

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: Taticca Auditores Independentes SS.
Auditor in the 2022 FCA: Taticca Auditores Independentes SS.
Auditor in the 2023 FCA: Taticca Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: DRS Auditores -> Michelon & Cia Auditores e Consultores SS. The outgoing firm audited 5 consecutive fiscal years (2007 to 2011), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2015: Michelon & Cia Auditores e Consultores SS -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples. The outgoing firm audited 3 consecutive fiscal years (2012 to 2014), by the start and end dates in the FCA.
- 2020: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples -> Taticca Auditores Independentes SS. The outgoing firm audited 5 consecutive fiscal years (2015 to 2019), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-11T06:40:17+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: RECRUSUL SA.

## Annex: full list of differences in the following year's comparative (R$ million)

(b) BPA 1.02.01.06 Tributos Diferidos: 0.0 in the version used, 2.0 in the comparative (difference of 2.0)
(b) BPA 1.02.01.06.03 Depósitos Judiciais: 0.0 in the version used, 2.0 in the comparative (difference of 2.0)
(b) BPA 1.02.01.09 Outros Ativos Não Circulantes: 0.0 in the version used, 1.0 in the comparative (difference of 1.0)
(b) BPA 1.02.01.09.03 Creditos à Realizar C/Ativos: 0.0 in the version used, 1.0 in the comparative (difference of 1.0)